

The Constructor-Theoretic Foundation of Mathematical Reality

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Abstract

Constructor theory expresses physical laws in terms of which transformations are possible and which are impossible, rather than as trajectories from initial conditions. I argue that this position finds its deepest justification in the nature of mathematical structure itself. The pattern-randomness dichotomy (that all existence is either patterned or not-patterned (random), both of which are mathematical) is itself a constructor-theoretic principle: it specifies which substrates for physical law are possible (mathematical structures) and which are impossible (non-mathematical “stuff”). This grounds constructor theory’s counterfactuals in logical necessity rather than empirical generalization. Furthermore, knowledge (which Deutsch and Marletto identify as a central constructor-theoretic concept) emerges naturally as self-referential mathematical structure that causes itself to remain instantiated. The hard problem of consciousness dissolves: experience is what certain mathematical structures are from the inside. Constructor theory and mathematical monism are mutually reinforcing: constructor theory provides the language, mathematical monism provides the ontology.

Keywords: constructor theory, counterfactual physics, Deutsch, Marletto, mathematical monism, pattern-randomness dichotomy, task ontology, substrate-independence, knowledge, hard problem of consciousness

1. Constructor Theory’s Promise

Constructor theory, developed by Deutsch and Marletto, proposes a fundamental reorientation of physics. Rather than expressing laws as differential equations governing trajectories from initial conditions, constructor theory expresses laws as principles about which physical transformations (tasks) are possible and which are impossible.

This goes beyond reformulation. Constructor theory can express laws that the trajectory-based approach cannot, including:

- The principles of information (what can and cannot be copied, transmitted, computed)
- The substrate-independence of computation

- The second law of thermodynamics in exact form
- The nature of knowledge as a physical phenomenon

The framework places counterfactuals, statements about what could happen, at the foundation of physics. A task is possible if no law forbids it; impossible otherwise.

But what grounds these counterfactuals? Why are certain transformations possible and others impossible? Constructor theory, as currently formulated, takes certain principles as given. I propose that mathematical monism provides the deeper foundation.

2. The Dichotomy as Constructor-Theoretic Principle

Consider the following principle:

The Substrate Principle: The only possible substrates for physical law are mathematical structures. Non-mathematical “stuff” is impossible.

In constructor-theoretic notation:

- Task T_{math} = “instantiate physics in mathematical structure” $\rightarrow$ **Possible**
- Task T_{nonmath} = “instantiate physics in non-mathematical substrate” $\rightarrow$ **Impossible**

This is a constructor-theoretic claim, it specifies what is possible and what is impossible. But unlike empirical constructor-theoretic principles, it follows from logical necessity.

The argument:

1. All existence is either patterned or not-patterned (random)
2. Patterns are mathematical structures by definition
3. Randomness is characterized by algorithmic incompressibility (high Kolmogorov complexity), itself a mathematical property
4. There is no third category
5. Therefore, whatever exists is mathematical structure

This is a logical partition. “Patterned or not-patterned” is exhaustive in the way “A or not-A” is exhaustive. Any purported non-mathematical substrate must either have structure (hence patterned, hence mathematical) or lack it

(hence random and still mathematically characterizable). No third ontological category remains.

The Substrate Principle thus has a special status among constructor-theoretic principles: it is necessarily true. It constrains all possible physics, all possible physics. In Deutsch’s terminology, it is a “superprinciple” (a principle about which principles are possible).

3. Why This Matters for Constructor Theory

Constructor theory currently takes certain principles as given, the interoperability principle, the composition principle, the principle of testability. These are proposed as deep laws, but their status remains somewhat unclear. Are they empirical generalizations? Necessary truths? Methodological commitments?

Mathematical monism clarifies this. If reality is mathematical structure, then:

Constructor-theoretic principles are truths about the space of mathematical structures.

A task is possible if it can be instantiated in some mathematical structure; impossible if no consistent mathematical structure can instantiate it. The counterfactuals of constructor theory are grounded in mathematical consistency.

The laws *are* the things. There is no separate realm of rules governing a realm of objects. The rules and the objects are both structure, the same structure, described differently. Constructor theory, by focusing on possible and impossible transformations, captures this unity: what can happen *is* what the structure is.

This explains why constructor theory can express laws that trajectory-based physics cannot. Trajectory-based physics describes the behavior of one structure. Constructor theory describes the space of possible structures, which transformations any structure could support.

4. Knowledge as Self-Referential Structure

Deutsch emphasizes that knowledge is a physical phenomenon with a constructor-theoretic characterization: knowledge is information that, once physically instantiated, tends to cause itself to remain instantiated. It survives criticism, testing, and error-correction. In Popper’s terms, knowledge is conjectural but non-arbitrary, it has been tested against reality and not yet refuted.

Mathematical monism deepens this account. Knowledge is self-referential mathematical structure, patterns that model themselves and their environment, and

in doing so, preserve themselves. The Popperian process of conjecture and refutation is itself mathematical: conjectures are structures, refutations are demonstrations of inconsistency with other structures (observations).

This connects to the nature of consciousness. Deutsch has noted that constructor theory places knowledge “center stage” in physics, but the question of subjective experience, what it is like to be a knowing system, remains open.

Mathematical monism provides the answer: consciousness is what self-referential mathematical structure is from the inside. There is no gap between structure and experience because structure, sufficiently self-referential, constitutes experience. A system that models itself *is* a perspective, constitutively.

The hard problem of consciousness dissolves, by recognizing that mathematical structure, when it models itself, is experience. The “inside view” is not added to structure; it is an aspect of what complex self-referential structure is. This is a recognition of what the concepts involved actually mean.

5. The Constructor Theory of Existence

Combining constructor theory with mathematical monism yields a result. Consider the question: which mathematical structures exist?

The constructor-theoretic answer: **A structure exists if and only if it is self-consistent.** Existence is not a task that must be performed; it is the default state of consistent structure. Non-existence would require a “filter”, something that prevents consistent structures from existing. But any such filter would itself be a mathematical structure, and would exist. The null filter (which excludes nothing) exists. Therefore, all consistent structures exist.

This is Tegmark’s Level IV multiverse, but grounded rather than postulated. And it is naturally expressed in constructor-theoretic terms:

Existence Principle: For any self-consistent mathematical structure S , instantiation of S is a possible task. No law of physics forbids consistent structures from existing. Conversely, no process can destroy mathematical consistency: the task of making $2+2$ cease to equal 4 is impossible in the constructor-theoretic sense.

The “task” here is abstract, existence is not caused by a constructor. But the constructor-theoretic framing clarifies what’s at stake: existence is possible for all consistent structures, impossible for inconsistent ones.

A clarification is needed. One can *describe* an inconsistent structure: the string “square circle” exists, and the brain state entertaining it exists. These descriptions are themselves consistent structures, and every medium carrying them (physical inscription, neural encoding) is a consistent mathematical structure, which is the sole criterion for existence. But the task of *instantiating* what

they describe is impossible: no consistent structure realizes a square circle under standard geometric definitions. To make the description refer to something that exists requires redefining one or more terms, and that redefinition is itself a consistent construction. In constructor-theoretic language: the task of rescuing an inconsistency is always possible, but its constructor is consistent reasoning. The only building material is reason. This is not a stipulation but a structural fact about what construction requires.

6. Testability and Explanation

Deutsch insists that scientific theories must be explanatory, not merely predictive. Constructor theory extends what can be explained by allowing counterfactual principles into physics.

Mathematical monism extends this further. It explains:

Why constructor theory works: Because reality is mathematical structure, and constructor theory is the natural language for describing the space of possible structures.

Why knowledge exists: Self-referential structures that model their environment and themselves have a tendency to persist, this is a mathematical fact about certain kinds of patterns.

Why the laws of physics take the form they do: The laws we observe are those compatible with the existence of observers. Observers are self-modeling structures. The laws must permit such structures.

Why anything exists at all: Existence requires no cause; consistent mathematical structure simply is. The question “why is there something rather than nothing?” dissolves, “nothing” (the absence of all structure) is itself a (trivial) mathematical structure, and cannot exclude other structures from existing.

These explanations are not probabilistic. They do not say “this is likely” but “this follows necessarily given the nature of mathematical structure.”

6.1 Structural Consequences

Constructor theory takes its impossibility principles as physically contingent: the universe happens to obey them. Mathematical monism predicts they are necessary: any consistent structure capable of supporting observers must obey them. This yields a testable distinction.

Consequence 1: Impossibility principles are necessary, not contingent. Constructor theory’s information principles state that certain transformations are possible (information can be copied, transmitted, computed) and others impossible (perfect cloning of unknown quantum states). Under constructor theory alone, these are contingent features of our physics. Under

matheism, they are structural necessities: any consistent mathematical structure rich enough to contain self-modeling subsystems must support information copying, because self-modeling requires internal representation, and representation is copying. The impossibility of perfect quantum cloning follows from unitarity, which itself has lower K than non-unitary alternatives. The structural consequence: no consistent mathematical structure supporting observers violates constructor theory's core information principles. If a counterexample is found (a consistent structure with observers but without information-copying), the consequence fails.

Consequence 2: Constructors have a logical direction that unitarity does not capture. Deutsch treats reversibility as fundamental: unitary quantum evolution is reversible, and quantum computation is built on reversible gates. Bennett (1973) showed that computation can be made thermodynamically reversible. But computation has a logical direction independent of thermodynamics. The simplest example: $1+1=2$ is deterministic forward. $2=x+y$ has infinitely many solutions backward. This is not a property of addition specifically. Nearly all computations are many-to-one functions: multiple inputs map to the same output. The forward direction is determined; the backward direction is not. This applies to constructors: any constructor that performs a many-to-one transformation has a logical direction, even if the transformation is thermodynamically reversible and the underlying dynamics are unitary.

This creates a genuine tension within constructor theory. Deutsch places reversibility at the foundation of physics. But his constructors compute, and computation has a direction. A constructor that performs a task, any task involving many-to-one mapping, has a before and an after that are not interchangeable. The constructor can be re-used (that is what makes it a constructor), but each use has a logical arrow. The arrow of computation is not the thermodynamic arrow. It is deeper: it follows from the logical structure of functions, not from entropy.

The standard response: "The function is many-to-one but the underlying physics is one-to-one (unitary). The arrow is in the description, not in reality." Under mathematical monism, this escape does not hold. There is no underlying physics separate from the mathematical structure. The function is not a description of something deeper. It is the thing. The unitarity of the wavefunction is one mathematical fact. The many-to-one-ness of the computed function is another. Both are equally real, equally fundamental, equally the physics. Deutsch privileges unitarity as the deeper level and calls the function coarse-grained. On what grounds? Both are mathematical structure. Neither is more real than the other. The arrow in the function is as fundamental as the reversibility in the wavefunction. Reality contains both a reversible substrate and irreversible functions operating within it. The physics cannot reverse the function because the physics IS the function.

The point is stronger than information loss. The inverse function does not exist. $3+4 \rightarrow 7$ is a computation. $7 \rightarrow (3,4)$ is not. It is an equation with infinitely many

solutions, not a function. There is nothing to compute in the backward direction. Not something hard to compute. Not something that loses information. Nothing. The backward direction is the nonexistence of a function.

Reversible computing (Bennett, 1973) does not change this. It computes a *different* function: $(3,4) \rightarrow (3,4,7)$, a bijection that stores inputs alongside output. The inverse of this different function does exist. But the original function's inverse still does not. The workaround is a new computation, not the reversal of the old one. Deutsch's reversible quantum gates are bijections by construction. They implement the log-keeping function, not the original many-to-one function. The arrow in the original function is untouched. Without consciousness, time is just another math dimension in the block universe. The arrow is not a feature consciousness observes; it is what consciousness is.

Consequence 2: The arrow of computation is structural, not perspectival.

A concrete walkthrough makes the argument precise. Consider classical addition: $f(1,1) = 2$. This function is many-to-one: $f(0,2) = 2$ also. The inverse $f^{-1}(2)$ is not defined: it is not a hard computation, it is not a lossy computation, it is not a computation at all. $2=x+y$ is an equation with infinitely many solutions, not a function. There is nothing to compute in the backward direction.

Deutsch's reversible version: define $g(a, b, 0) = (a, b, a+b)$. So $g(1, 1, 0) = (1, 1, 2)$. This IS bijective: $g^{-1}(1, 1, 2) = (1, 1, 0)$. The inverse exists. Has the computation been reversed? No. g is a different function from f . g stores the inputs alongside the output. g^{-1} is the inverse of g . It is not the inverse of f . f still has no inverse. Computing g instead of f does not make f reversible. It changes the subject.

In MWI, the total quantum state evolves unitarily (one-to-one). Nothing is ever discarded. The universe computes g , not f . An observer in one branch sees 2 and loses track of the inputs, but the global wavefunction retains them. Deutsch's conclusion: the many-to-one structure is perspectival, an artifact of tracing out part of the system. The physics is always one-to-one. The arrow is in the description, not in reality.

Under mathematical monism, this escape does not hold. Both f (many-to-one) and g (one-to-one) are mathematical structures. Both exist. Both are equally real. Deutsch privileges g as "the real physics" and calls f a coarse-grained approximation. On what grounds? Both are mathematical structure. The lack of an inverse for f is a structural fact. Computing g instead does not make f invertible, any more than the existence of rectangles gives squares equal diagonals. f 's arrow is absolute, present in every consistent structure that contains many-to-one functions, which is nearly all of them.

Constructor theory is itself a framework of local operations. It defines physics through which transformations substrates can undergo. Deutsch appeals to global unitarity to dissolve the arrow. That appeal abandons the level of de-

scription his own framework selects. If constructor theory’s local operations are the right vocabulary for physics, the arrow those operations contain is physical.

Timpson (2005) analyzed the Deutsch-Hayden formalism and its claim to eliminate nonlocality through purely local information flow. His analysis showed that whether information is truly “local” in the Deutsch-Hayden picture depends on what counts as physically significant information, a question the formalism alone does not settle. The same ambiguity arises here. Deutsch appeals to the global wavefunction as the physically significant level of description, rendering many-to-one functions perspectival. Timpson’s observation supports the present argument: the choice between local and global descriptions is not settled by the formalism. Under mathematical monism, both descriptions are equally real mathematical structures. By the framework Deutsch himself endorses, constructor theory defines physics through local transformations, and those transformations contain the arrow.

This connects to constructor theory’s own defining concept. A constructor performs a task and returns to its ready state. The ready state is the same regardless of which specific task instance just ran. The constructor sheds all information about what it just did. This reset IS many-to-one: multiple distinct task-executions map to the same ready state. If the constructor keeps all ancilla bits to preserve reversibility, it never returns to its ready state. A constructor that does not return to its ready state is not a constructor. Reversibility or constructors. Not both. The logical arrow is built into the constructor primitive.

6.2 How This Strengthens Many-Worlds

The finding does not weaken MWI. It strengthens it by removing a burden.

Under collapse interpretations, the wavefunction itself is many-to-one: superposition collapses to one outcome, information is destroyed at the substrate level. The arrow exists at both the function level and the substrate level. Under SDP, this is expensive: specifying which outcome was selected at each measurement requires unbounded K .

Under MWI, the substrate remains one-to-one (unitary). Information is preserved globally. The arrow exists only at the function level, where it is a structural fact about many-to-one mappings, requiring no additional postulate. MWI is the only interpretation where the substrate is clean and the arrow needs no thermodynamic or collapse-based explanation. The arrow comes from the mathematics of functions, not from the physics of collapse.

This means MWI gives the cleanest architecture: reversible substrate plus irreversible functions. Collapse interpretations are overkill. They break reversibility at the substrate level when the arrow already exists at the function level. SDP favors MWI because it achieves both (global reversibility, local arrows) with lower total K than any interpretation that adds collapse.

Consequence 3: SDP selects among constructor-theoretic frame-

works. Among all possible sets of impossibility principles compatible with observers, ours is the simplest (lowest K). SDP predicts that discovering our physics has more impossibility principles than minimally necessary to support observers would challenge the framework.

7. What This Framework Does Not Claim

To avoid misunderstanding:

No probability over MWI branches: This framework does not assign probabilities to branches of the quantum multiverse. The Born rule is a separate question, which Deutsch has addressed through decision theory. Mathematical monism is compatible with that approach.

No consequence for specific laws: Mathematical monism does not predict the Standard Model. It explains why there are laws at all, and why they take constructor-theoretic form. The specific laws are empirical.

No replacement of constructor theory: Mathematical monism provides a foundation for constructor theory, not a replacement. Constructor theory remains the appropriate language for expressing physical laws.

8. Objections

“This is philosophy, not physics.”

Constructor theory itself straddles this boundary. It expresses principles that constrain all possible physics. Mathematical monism does the same at a deeper level. The distinction between physics and philosophy is less clear than often assumed, as Deutsch has noted, having the right philosophy is crucial for physics.

“Mathematical monism is not testable.”

It makes the structural wager that constructor theory will succeed, that all fundamental laws can be expressed in constructor-theoretic form. If some law resisted such expression, mathematical monism would be challenged. More specifically: mathematical monism entails that non-computable processes have no physical effects. If physics required essential non-computability, this would be evidence against mathematical monism.

“How can structure be experience?”

The question assumes that structure and experience are different kinds of things that must be connected. Mathematical monism denies this assumption. Self-referential structure does not “generate” experience; it is experience, viewed from the inside. The demand for an explanation of how structure becomes experience presupposes the dualism that mathematical monism rejects.

9. Conclusion

Constructor theory and mathematical monism are natural partners. Constructor theory provides the language for expressing physical law, possible and impossible transformations. Mathematical monism provides the ontology, reality is mathematical structure, and constructor-theoretic principles are truths about the space of such structures.

Together, they offer:

- A foundation for constructor theory’s counterfactuals in logical necessity
- An explanation of knowledge as self-referential structure
- A dissolution of the hard problem of consciousness
- A grounding for the existence of all consistent mathematical structures

Physics, properly understood, is the study of which transformations are possible within mathematical structure. Constructor theory makes this explicit. Mathematical monism explains why this is so.

The rules *are* the things. The math *is* the territory. One plus one leads only to two, but two does not lead back to one plus one. Each timeline lacks a determined backward path, even if the multiverse globally stores all paths. Deutsch built a theory of physics around constructors. Constructors contain the arrow. The framework was always pointing at mathematical monism.

A methodological observation: none of these results required advanced mathematics. They required removing a single ontological assumption, that there exists something other than mathematical structure, after which the conclusions follow from elementary logic. The assumption functioned as a constraint on inquiry rather than a product of it. The constructor-reset observation, the irreversibility of many-to-one functions, the tension between unitarity and task-level directionality: each was available to anyone who examined the definitions without the prior commitment to a non-mathematical substrate. The difficulty was not in the reasoning but in seeing the box.

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